

Detection of Evil Flies: Securing Air-Ground Aviation Communication

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ABSTRACT

The aviation community is employing various air traffic control and mobile communication technologies, such as ubiquitous data links, wireless communication architectures and protocols. Recently, software-defined networking (SDN) based architectures (i.e., cockpit network communications environment testing (COMET)) have been proposed for Air-Ground communication. However, an evil can break the communication between a pilot and air traffic control, resulting in a hazardous (or life-threatening) situation up in the air or failure of ground equipment. This paper proposes an efficient evil detection and prevention mechanism (called DoEF) for the COMET architecture. The proposed DoEF utilizes a deep learning-based approach, i.e., long-short term memory (LSTM), to detect the evil flies and provide possible countermeasures. Our preliminary results show that the proposed scheme reduces the detection time and increases the detection accuracy of distributed denial of service (DDoS) attacks for the aviation network.

CCS CONCEPTS

• **Security and privacy** → Intrusion/anomaly.

KEYWORDS

Security, Intrusion detection, Aviation Network

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1 INTRODUCTION

According to Eurocontrol, air traffic is expected to double by 2025 in Europe, with an average growth of 2.7-3.7% every year. With the

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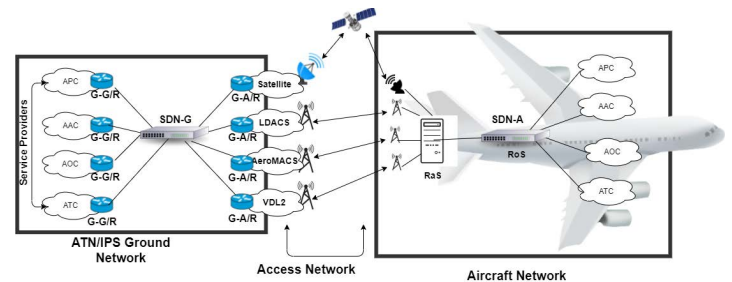


Figure 1: COMET Architecture – we have adopted the architecture from [4]

increased airspace usage, many aerospace companies have been employing various innovative solutions (e.g., network softwarization, cellularization, and so on) to manage air-ground communication. In this regard, the European aviation network, along with various research groups, have started several initiatives, for instance, Single European Sky ATM Research (SESAR) and CleanSky programs. However, the main focus of these initiatives is the digital transformation and de-carbonization for aviation communication through Internet protocols (IPs) and network softwarization. One of the popular architectures is Future Cockpit Network Communications Environment Testing (COMET), which exploits various wireless technologies and software-defined networking to manage aeronautical communication networks for air traffic management, shown in Fig. 1. One main purpose of the COMET is to design, estimate and evaluate the performance between air and ground communications. As shown in Fig. 1, the COMET architecture utilizes the software-defined network (SDN) controllers (for air (SDN-A) and for ground (SDN-G)). These controllers perform or operate control and management-related tasks (e.g., airline administrative control (AAC), air operation control (AOC), air traffic control (ATC), etc.).

Indeed, SDN supports various innovations, e.g., centralized management of control systems, simplified packet forwarding, efficient configurations, and so on. However, it opens an immense threat vector. For instance, a malicious attacker can trigger various irreversible distributed denial of service (DDoS) threats to the operating systems between the ATC and aircraft, as discussed in [2]. A report stated that Polish Airlines had targeted a massive malicious DDoS attack in 2015. This attack caused various operations halted up in the air, particularly, flight plan IT system [1]. As a result, such

innovative softwarizations may lead to catastrophic consequences to the individuals and airplanes (up in the air) and so to the ground network. Hence, there is an immense need for an efficient solution that can systematically detect the potential threat, such as DDoS, in the air-ground aviation networks.

In this paper, we propose a new model (called "detection of evil flies (DoEF)") in the air-ground communication with high accuracy. The main focus of DoEF is to propose a deep learning-based scheme (i.e., long-short term memory (LSTM)) that can detect evil flies and can protect the COMET architecture, mainly its SDN-controllers (i.e., SDN-G and SDN-A) from the DDoS attacks. We exploit the NSL-KDD dataset. Our preliminary results show that the proposed scheme not only reduces the detection time but increases the accuracy of distributed denial of service (DDoS) attacks for the aviation network. In addition, the proposed DoEF outperforms (using NSL-KDD) the existing work.

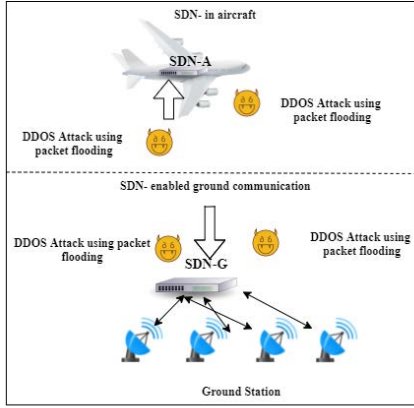


Figure 2: Threat Model for COMET architecture

2 PROPOSED DOEF ARCHITECTURE

As shown in Fig. 2, the SDN-A controls the airplane systems, and the SDN-G controls the ground system. We assume that an adversary's goal will be to launch the DDoS attacks to either SDN-A or SDN-G (or both) and interrupt the operations. An attacker can typically overwhelm the ATC (at the ground) with a massive amount of malicious traffic or packet flooding. Likewise, he/she can disturb the operation systems up in the air (airplane). In addition, the most dangerous aspect of DDoS attacks is that they consume all of the network's resources either up in the sky or on the ground when they are launched successfully, leaving no room for normal application traffic. This causes a full (or partial) communication breakdown, which is both disastrous and life-threatening, especially in avionics communications systems.

This section proposes first the detection of evil flies either up in the sky or in the ground (or both). To detect DDoS attacks, we used Long Short-Term Memory (LSTM), a deep learning-based approach. The motivation for selecting LSTM as a model is our dataset, i.e., time-series data. It is well known that LSTM performs very well on time series data. However, to detect evil flies, we have used the following steps:

Data Pre-processing: We used the NSL-KDD [3] dataset to train and test the proposed DoEF against normal and malicious traffic. NSL-KDD is the state-of-the-art dataset used for network-based intrusion detection systems (NIDS). After getting the NSL-KDD dataset, some pre-processing steps are performed as shown in Algorithm 1 (i.e., 1-4 steps). We normalized the dataset using a min-max normalization approach. NSL-KDD dataset has some non-numeric values (e.g., service, protocol type, flag, and class label), and deep learning performs well on non-numeric data. Therefore we perform binary data-encoding. We used 70% dataset for training, and 30% dataset is used for testing purposes.

LSTM Architecture Designing: In order to detect evil flies, we used LSTM model that consists of mainly three parts: (i) Input layer which consists of 41 features. These features are fed to hidden layers. We have only two classes, the normal class and the anomaly class. Therefore we will have only one output layer and one output (e.g., Normal traffic or DDoS attack). (ii) Hidden layers of our proposed LSTM architecture consist of 3 LSTM layers, three dense layers, and one dropout function to avoid over-fitting. We used sigmoid as an activation function. The role of the sigmoid function in LSTM is to bring non-linearity to the model. In simple words, it decides which value to pass to the output. (iii) the hidden layers compute the weighted inputs and generate net input, which is then combined with activation functions to produce the final output. Algorithm 1 (i.e., 5-13 steps) shows the LSTM-based DoEF. Finally, the overall structure of the proposed LSTM architecture is depicted in Fig. 3.

Algorithm 1: DoEF Algorithm

1. **Input:** data $\leftarrow$ NSL-KDD 41 features
 2. **Output:** $\leftarrow$ Normal Traffic or DDoS Attack
 3. $C \leftarrow E(\text{data})$ Data Encoding
 4. $C' = \frac{C - \min(C)}{\max(C) - \min(C)}$ Data Normalization
 5. $O \leftarrow LSTM(C')$ LSTM Base Model
 6. $PD \leftarrow \text{Predict Class}(O)$
 7. **For** i in range(1, len(PD)) **do**
 8. **IF** $(PD[i] = y \text{ test}[i])$ **then return** $PD[i]$
 9. **ELSE return** $y \text{ test}[i]$
 10. **End IF, End For**
 11. **For** each epoch in range (10): **do**
 12. Evaluate Loss, Validation Loss, Accuracy and Validation Accuracy, Precision, Recall and FAR
 13. **End For, return Output**
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3 EXPERIMENT SETUP AND RESULT

Google Colab, Google's online Graphical Processing Unit (GPU), is used for the experiment. We used Python 3.7 as our programming language and personal computer with a higher-capacity operating system (Windows 10), 8GB RAM, and a Core i7 system with a 1.8GHz processor for this study.

Our preliminary results are shown in Table 1, which represents the training loss, testing loss and accuracy, precision, recall, and time complexity of the proposed DoEF solution. We used 20 epochs for the training and testing. We achieved 9.84% training loss and

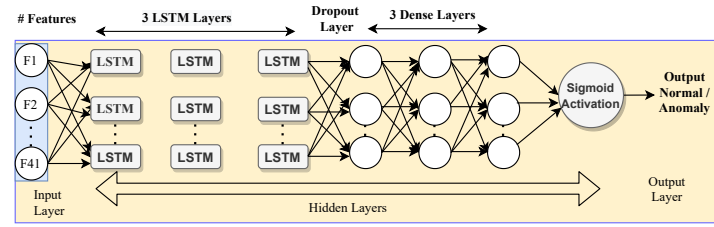


Figure 3: Proposed LSTM Architecture: 41 features are fed to the three Hidden Layers, consisting of three LSTMs, three dense layers, a dropout layer, and a sigmoid activation function. Finally, the output layer will detect normal/anomaly in the proposed architecture.

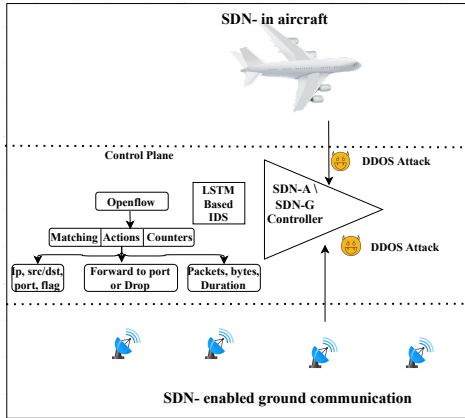


Figure 4: Countermeasure for the COMET architecture

Table 1: Evaluation Summary of Proposed DoEF

Metrics	ANN [2]	Proposed DoEF
Training Loss	10.18	9.84
Testing Loss	17.79	10.68
Training Accuracy	90.14	95.82
Testing Accuracy	76.28	95.81
Precision	85	95.50
Recall	76	96
Training Time (Sec)	-	519.60
Testing Time (Sec)	-	6.63
FAR	-	5.39

10.68% testing loss, respectively. Similarly, we achieved 95.82% training accuracy and 95.81% testing accuracy, respectively. In comparison with [2] authors used artificial neural network (ANN) to detect DDoS attacks in COMET. They achieved a 10.18% training loss, 17.79% testing loss, 90.14% training accuracy, and 76.28% testing accuracy. On the contrary, the proposed DoEF outperforms in terms of loss and accuracy compared with existing research. Our proposed solution also achieved a 10.50% high precision and 20% high recall score as compared with [2]. It is worth to note that the training and testing timings and false alarm rate are the most paramount components of machine learning-based systems. However, the DoEF requires 519.60 seconds for training and 6.63 seconds for testing. In

addition, the FAR of the DOEF is 5.39%. Whereas the training and testing timing and FAR in [2], neither considered nor mentioned. Hence, the proposed DoEF achieved high efficiency and reduced the detection time of evil flies.

4 COUNTERMEASURES AGAINST EVIL FLIES

To protect the COMET architecture, OpenFlow can play a very important role to detect the evil flies up in the air and on the ground (or both). The OpenFlow protocol instructs the corresponding switch to restrict/block a specific source of malicious traffic. For instance, as shown in Fig. 4, assume that an attacker sends malicious packets to the SDN controller, then the controller forwards these packets to the OpenFlow. Based on the policies, the OpenFlow decides whether to forward the packet to port A or drop. Note that the OpenFlow protocol is very successful in rapidly mitigating network anomalies via the use of flow tables. To prevent future attacks, the updated security policies and forwarding rules may be implemented on other switches in the network.

5 CONCLUSION

We proposed an efficient evil detection and prevention mechanism (called DoEF) for the COMET architecture. The proposed DoEF utilizes a deep learning-based approach, i.e., LSTM. We used 3 LSTM layers, three dense layers, one dropout layer, and a sigmoid activation function with a batch size of 32 and 20 epochs. Our proposed solution can detect DDoS attacks with 95.81% accuracy, 95.50% precision, 96% recall, and 5.39% FAR compared to other existing work. To protect such systems, the open-flow controller can be used to restrict/block after malicious traffic is detected successfully.

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